Real-time keyword spotting on ARM Cortex-M4F

Build and test

Continuous single-keyword spotting on an **Arduino Nano 33 BLE / nRF52840** (64 MHz Cortex-M4F). Interrupt-driven PDM acquisition feeds CMSIS-DSP feature extraction and an int8 TensorFlow Lite Micro model in a dedicated audio task. A second task filters queued scores and controls the device. Confirmed **"yes"** detections toggle the built-in LED. The application runs continuously while powered.

The engineering work covers interrupt/task synchronization, bounded buffering, CMSIS packed-FFT correctness, int8 inference, and detection filtering across queue overloads. Two application tasks separate audio processing from device actions; task separation uses preemption on one core rather than parallel CPUs. The primary workflow is command-line Make and C++; Arduino IDE is optional.

For a review of the implementation, start with the engineering overview: requirements, design tradeoffs, source entry points, and a mapping from claims to test evidence. The RTOS design explains synchronization and failure behavior; RTOS validation records the hardware results.

Measured results

Measurements below are from the two-task RTOS firmware on the connected Nano, recorded October 20, 2026. Timing summarizes normal-operation checks; the deliberate stall is documented separately. Memory is from the current object-based C++ firmware build. These are observed results, not worst-case guarantees.

Measurement	Recorded result
Audio rate / hop	16 kHz / 256 samples (16 ms)
FFT frame / overlap	512 samples (32 ms) / 50%
Model input	61 x 20 int8 log-mel features (~1 s context)
Decision cadence after warmup	Every 8 feature hops (~128 ms), when audio backlog is drained
DSP average / observed maximum	2.63 / 2.86 ms
Inference average / observed maximum	59.13 / 59.65 ms
Processing iteration observed maximum	62.23 ms
Circular audio buffer / observed peak occupancy	8192 / 768 samples
Flash / static RAM	331,936 / 209,320 bytes (33% / 79%)
Dedicated task stacks	Audio 8192 bytes; control 6144 bytes
RTOS spoken "yes" check	11 confirmed detections; physical LED toggling confirmed
Reported sample drops / missed buffers / inference errors	0 / 0 / 0 in the recorded checks
Normal-operation score drops / stale results	0 / 0
Forced 1.5 s control stall	4 score drops; 9 stale results rejected; 0 reported audio losses

See RTOS validation for current task, stress, and speech-test evidence. The earlier module/refactor record and pre-RTOS build check retain historical results. Response measurements start at a high-confidence window's estimated

end; they do not measure acoustic word-onset latency. Detection accuracy across a controlled speaker/noise dataset and hours-to-days endurance have not been established.

Measured memory margins

These high-water measurements come from the object-refactor normal and stress checks. They describe exercised paths, rather than guaranteed worst-case usage.

Allocation	Capacity	Observed peak use / remaining
Audio-task stack	8192 bytes	1416 used / 6776 unused (83%)
Control-task stack	6144 bytes	1168 used / 4976 unused (81%)
Audio ring	8192 samples	768 queued / 7424 spare (91%)

At peak observed ring occupancy, the remaining capacity represents about 464 ms of additional audio at 16 kHz. This is buffer capacity, not permission to block audio processing indefinitely. The eight-score control queue intentionally overflows in the 1.5 s stall test; dropping scores and rejecting stale evidence preserves acquisition but can miss a command during the stall.

The static RAM total already includes both application stacks and the 96 KiB tensor arena. The build leaves 52,824 bytes outside static allocations; this is not measured free heap. Peak runtime heap, framework/interrupt-stack usage, and unused tensor-arena capacity have not been measured. Broader memory headroom must be established before substantial expansion.

Build, test, and flash

Prerequisites: **Python 3.9+**, GNU Make, and a host C++ compiler with AddressSanitizer/UndefinedBehaviorSanitizer support. Supported firmware-build hosts: macOS Intel/Apple Silicon and Linux x86-64/ARM64. Network access is needed for initial setup. Apple Silicon also needs Rosetta 2 for the pinned Intel Arm compiler. No Arduino IDE or globally installed Arduino CLI is required.

```
git clone https://github.com/jameslavrenz/KWS_Arm_CortexM4F.git
cd KWS_Arm_CortexM4F
make setup                          # Download and verify pinned tools/libraries
make test                           # Host unit and integration tests
make build                          # Cross-compile for Nano 33 BLE
make boards                         # Find the connected board's serial port
make flash PORT=/dev/cu.usbmodem11201  # Example macOS port; use your board's port
# Linux example: make flash PORT=/dev/ttyACM0
```

`make build` writes ELF/BIN/HEX outputs and a source/dependency manifest to `.build/artifacts/`. `make flash` rebuilds before uploading. All downloaded tools, board packages, libraries, generated sketch files, and objects stay under the ignored `.build/` directory. Existing Arduino IDE libraries are not used. Use `JOBS=4` to change firmware compile parallelism. A board is needed only for `make boards`, flashing, and physical tests; host tests and compilation can run without one. `make help` lists the available targets.

For prerequisite installation, expected output files, serial-port examples, troubleshooting, and rebuilding the PDF guide, follow the complete build guide.

dependencies.lock.json pins Arduino CLI 1.5.1, Arduino mbed Nano core 4.6.0 (GCC Arm Embedded 7-2017q4), CMSIS-DSP 5.7.0, and TensorFlow Lite Micro's exact source revision. CLI/library downloads are SHA-256 checked; Arduino CLI verifies core/tool package downloads. The three required TFLite compatibility headers are tracked in `build-support/`. See build reproducibility for dependency provenance and limitations.

GitHub Actions runs **the same** `make test` **and** `make setup && make build` **paths** on pushes and pull requests, plus `make format-check` with pinned formatters. Firmware and build manifests are retained as CI artifacts. Physical microphone/LED testing remains a hardware check.

Source language and formatting

All project-owned firmware and native tests are **C++**, including the minimal `.ino` entry point and `.h` declarations; there are no project-owned `.c` translation units. Host tests compile with C++11; the pinned board core compiles firmware with GNU++14. Python supplies host tooling/training, and a shell script runs the tests. The downloaded Arduino/Mbed and CMSIS libraries retain their upstream C/C++ implementations.

Maintained code uses **four spaces per indentation level**, with `.clang-format`, Black configuration, and `.editorconfig` checked into the repository. Makefile recipes retain their required tabs. Run `make format` to apply the style and `make format-check` to verify it; the formatting guide documents the project-local formatter setup. Native test fixtures are ordinary C++ source/header files under `tests/fixtures/`.

Runtime architecture and ownership

```
PDM / DMA -> interrupt -> circular audio buffer
Audio task: frame overlap / FFT / log-mel -> int8 model
  -> eight-entry RTOS value queue -> control task
Control task: confidence filter -> LED; USB telemetry
```

Module	Owned state and responsibility	Execution context
`audio_capture`	PDM staging, ring, acquisition counters, complete-hop reads	Interrupt producer; audio-task consumer
`audio_pipeline`	Hop processing, readiness, backlog deferral, DSP/inference timing	Audio task
`feature_extractor`	Hann/FFT workspace, overlap, quantization, rolling features	Audio task
`keyword_model`	Interpreter, 96 KiB tensor arena, class-0 probability	Audio task
`keyword_event` / `keyword_detector`	Event freshness, confidence filtering, cooldown, response evidence	Control task
`device_control`	Built-in LED state and confirmed-event action	Control task after initialization
`telemetry`	Output staging and nonblocking USB transmission	Control task after initialization
`kws_pipeline`	Task/queue startup, transport, generations, metrics synchronization	Startup plus two task entry points
`kws_config`	Shared dimensions and inference cadence	Compile-time configuration

Stateful modules use C++ objects: `FeatureExtractor` owns fixed-size DSP buffers, `AudioCapture` owns the microphone ring and callback context, and `DeviceControl` and `Telemetry` own their respective state. `AudioPipeline` receives its capture, extractor, and model dependencies by reference. A private application object owns the RTOS resources; the single-model `KeywordModel` encapsulates its fixed TFLite runtime. See C++ ownership and refactor validation.

The `.ino` only delegates startup and idles the existing Arduino main task. Application behavior lives in `.cpp`/`.h` modules. The Makefile invokes a pinned Arduino CLI, which configures the Arm cross-compiler and board framework; it still compiles the minimal `.ino`. This is an IDE-independent build using the Arduino Mbed core, not a standalone replacement for that framework. Two application tasks use the board's built-in CMSIS-RTOS2/RTX kernel: audio at `AboveNormal` priority with an 8 KiB stack, control at `Normal` with a 6 KiB stack. Stacks, thread control blocks, queue

storage, and mutex storage are static. The callback sets an ISR-safe thread flag after publishing audio. Flags may coalesce; the audio task drains complete hops before waiting again.

The queue copies timestamped score values, including low scores needed to rearm the detector. No audio buffers, feature windows, or tensor pointers cross tasks. A priority-inheriting mutex protects only a short metrics copy. The interrupt callback never takes that mutex. Audio never waits for room in the event queue. See RTOS design for scheduling, ownership, and overload rules.

DSP pipeline

The audio task scales signed PCM by `1/32768`, forms 512-sample frames with 256-sample overlap, applies a symmetric Hann window, and computes a CMSIS real FFT. Correct packed-bin handling produces 257 power values. A 20-band mel matrix (20-4000 Hz) projects the power spectrum; natural log with epsilon `1e-6` and model-provided int8 quantization produce each feature row. The rolling 61 x 20 input spans 992 ms of audio; inference runs nominally every eight new rows (128 ms) once ready and caught up.

The DSP pipeline guide documents stage equations, buffer dimensions, startup/reset behavior, model layout, timing boundaries, and an existing training/firmware Hann-window convention difference. Full numerical feature parity with the training notebook has not been established.

TinyML neural-network inference

The model is an int8 depthwise-separable CNN with input `[1,61,20,1]` and three softmax outputs: `yes`, unknown speech, and silence. The training model has **1315 Keras parameters** (1155 trainable); the converted graph contains **992 int8 weights and 83 int32 biases**, or 1075 parameter values after batch-normalization folding. Six convolutions, spatial mean pooling, and softmax form the eight-operator inference graph.

Roughly low-to-mid 90% keyword-recognition success during informal use with this 1075-parameter deployed model. This is an approximate observation under current operating conditions; a labeled trial count and false-trigger rate have not been supplied for the estimate. **An untested hypothesis is that a wider or deeper model, with suitable training and tuning, could improve recognition into the mid-to-high 90% range.** Additional parameters alone do not guarantee that improvement; a controlled comparison must also check false triggers, latency, and RAM use. See recognition evidence and hypothesis.

The **8496-byte model blob and its weights reside in internal flash**, compiled from the tracked header. Inference reserves a **96 KiB SRAM tensor arena** for activations/scratch/runtime metadata; the current profiler additionally occupies 20,488 bytes of static SRAM. These allocations are already included in the reported firmware memory totals. The TinyML inference guide documents layer counts, quantization, model identity, storage placement, runtime flow, inspection commands, and the limits of the recorded training accuracy.

Real-time behavior and design choices

Acquisition. The PDM callback drains bounded reads into a single-producer, single-consumer circular buffer using lock-free atomic publication. It does no DSP, inference, GPIO action, or USB printing. The audio task consumes complete 256-sample hops. The 8192-sample capacity provides 512 ms of nominal buffering at 16 kHz; it absorbs bounded stalls rather than sustained overload.

Scheduling. Capture starts after model/DSP initialization. The first decision waits for a full feature window; subsequent decisions require eight new feature frames and less than one queued hop. FFT frame overlap continues across model calls. DSP and inference leave interrupts enabled; short counter snapshots mask interrupts. With the recorded averages, eight DSP hops plus one inference take about 79 ms of a nominal 128 ms decision interval, excluding other work.

FFT correctness. CMSIS fast real FFT packs DC at output index 0 and Nyquist at index 1. `fft_power.h` treats both separately and computes interior-bin power without reading beyond the 512-float output. Tests exercise exact-size arrays under address/undefined-behavior sanitizers.

Detection. Class 0 is `yes`; the training labels also include unknown speech and silence. A confidence EMA with weight 0.5 must reach 0.75 while raw confidence is at least 0.8 on two successive decisions. The filter rearms below 0.4 and requires 1500 ms between triggers. Thresholds are starting values for broader speech/noise validation, rather than an established false-positive guarantee.

Failure behavior. A full ring rejects new samples, counts dropped samples and affected callbacks, and preserves unread samples. On observing a loss, the audio task discards pending audio, resets features, invalidates older queued scores, and waits for a fresh feature window. Failed inference increments an error counter and invalidates detection evidence. A full event queue drops a score and advances a generation counter rather than blocking audio. The control task rejects old generations and scores older than 250 ms; sequence/cadence gaps reset temporal evidence while retaining latch/cooldown history. Startup checks DSP/model/PDM and RTOS objects/tasks, cleaning up tasks on partial failure. Automatic recovery from peripheral failure is future work.

Tests and instrumentation

`make test-unit` runs nine component checks. `make test-regression` runs the complete thirteen-check suite, including those unit tests plus stream, pipeline, RTOS startup, and build-tool checks. `make test` runs the same complete regression gate in CI. See the testing guide for coverage, commands, hardware procedures, and limits.

The suite exercises queue ordering/overflow/wraparound, packed FFT bins/bounds, simulated inference stalls, temporal filtering and cooldown, detection-response timing and timer wraparound, actual GPIO-module logic with fake pins, and the actual audio pipeline with fake capture/features/model modules. New checks exercise the actual extractor against frozen pre-refactor feature vectors and the actual PDM receiver against a deterministic driver fake. RTOS startup tests check static storage, priority/stack configuration, callback notification, and cleanup after initialization failures. Event tests cover stale generations, queue gaps, duplicate scores, timestamp wraparound, and cooldown retention. Build-tool tests check archive checksums, extraction boundaries, library/lock drift, and environment isolation.

At 115200 baud, USB telemetry is emitted every two seconds without waiting for a host. A two-second startup timeout allows standalone operation.

Field	Meaning
`samples`, `dropped`, `missed_buffers`	Received sample count; ring loss; callbacks with loss
`queue_max`, `gap_max_us`, `audio_age_us`	Peak backlog; maximum callback gap; time since last callback
`dsp_us`, `infer_us`, `process_us`	Average/maximum execution times in microseconds
`response_us`	Last/maximum estimated high-confidence-window-to-action delay
`inferences`, `errors`, `p_milli`	Model calls, failures, class-0 confidence x 1000
`triggers`, `led`	Confirmed actions and current logical LED state
`event_drops`, `stale_events`, `notify_errors`	Nonblocking queue send failures, discarded stale scores, ISR flag errors
`event_us`	Average/maximum inference-result transport delay
`stack_free`	Minimum unused stack bytes: audio/control, from the RTX watermark

For a live capture, install `pyserial` in a host virtual environment and run:

```
python3 tests/hardware_capture.py --port /dev/cu.usbmodem11201 --seconds 60
```

Say "yes" with pauses of about three seconds and observe the LED. The script checks acquisition/inference progression and reported loss/error counters; physical LED behavior and speech-trigger correspondence require observation. The host recording ends after the requested interval; firmware keeps running.

Model and project artifacts

- Training notebook: Google Speech Commands training and quantization workflow.

- Model architecture: exported model visualization.

- Feature timing diagram: context and overlap illustration.

- Documentation index: initial and refactor validation records with raw telemetry.

- Engineering overview: design decisions, source navigation, and evidence for reviewers.

Next engineering work

Evaluate false accepts/rejects across speakers, noise, distance, and silence; run longer endurance tests; measure runtime heap, framework/interrupt stacks, and tensor-arena usage; add peripheral fault recovery; and extend the control task with communications or additional device functionality. Threshold refinement and multi-keyword training are separate from the build/runtime work.